

A Spectral Reduction of the Collatz Conjecture via Phantom Orbit Shadowing

(Version 2 – with Lean 4 verification and updated literature)

Piero Borgatta*
Independent Researcher

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Abstract

We propose a structural reformulation of the Collatz conjecture in terms of a finite-dimensional spectral problem. Starting from the empirical observation that “rebel” orbits of the Syracuse map shadow, for finite stretches, periodic rational points of the 2-adic dynamics associated to negative rational fixed points (*phantom cycles* in the terminology of [9], introduced concurrently and independently of this work; the underlying object is classical and goes back to [2]), we construct an episode graph on the space of phantom orbits. The strongly connected component (SCC) of this graph that supports the slowest descent is encoded as a finite weighted transfer operator on a refined phase state $(\nu_2(t), \text{odd}(t) \bmod 4, h \bmod 4)$ at the critical node ($k = 12, c = 2, b = 1$).

We prove an exact congruential shadowing lemma reducing the question of arbitrarily long shadowing episodes to a residue condition $n \equiv q_w \pmod{2^{bA+1}}$, where q_w is the 2-adic representative of a phantom word w and $A = |w|_{\nu_2}$ its period sum. The lemma is the iterated step-by-step form of the per-block 2-adic repulsion identity of [9, Prop. 7.4] and is equivalent in content to the classical inverse formula for parity vectors [13, Lemma 1]; the iterated form is what feeds the episode graph construction. **The lemma and its no-infinite-shadowing corollary have been mechanically verified in Lean 4 with Mathlib** (Section 3.3); the project is sorry-free.

We then build, for each truncation depth T and high-bit lift count j , a finite matrix $\text{FULL}_{T,j}$ encoding first-return behavior at the critical node, decompose it as $\text{FULL}_{T,j} = \text{CORE}_{T,j} + \text{TAIL}_{T,j}$, and apply a weighted Collatz–Wielandt bound [19] with respect to the right Perron eigenvector v of $\text{FULL}_{T,j}$. Our numerical results show

$$\rho(\text{FULL}_{T,j}) \leq \max_i (\text{CORE}_{T,j} v)_i / v_i + \max_i (\text{TAIL}_{T,j} v)_i / v_i \leq 0.052 \quad \text{for } T \leq 16 \text{ and } j \leq 32,$$

with monotone but geometrically decaying increments in both T and j , and we extend the bound to a cross-node operator on the SCC obtaining $\rho(M_{\text{cross}}) \leq 0.0366$. The cross-node spectrum is *lower* than the single-node bound at the critical vertex.

We do not claim a proof of the Collatz conjecture. We claim that the structural problem at the critical SCC is reduced to two explicit a priori estimates on finite matrices: the asymptotic boundedness of the weighted bound under refinement in T , and the closure of the empirical signature support under refinement in j . The same distributional-to-pointwise barrier identified in [9] manifests itself here in finite-matrix form (Section 9). We document supporting numerical evidence over a five-decade range of parameters, expose the open estimates as conjectures, and provide all computational scripts and the Lean 4 formalization as supplementary material.

*Independent Researcher, Italy. Email: info@pieroborgatta.com. ORCID: [0009-0001-8025-2405](https://orcid.org/0009-0001-8025-2405).

Research program note. This paper is the first artifact of a personal research program by the author, an independent researcher, exploring whether iterative collaboration with large language model assistants can support non-standard attempts at open mathematical problems. *Verifying the practical limits of such collaboration is an explicit secondary goal of this work, on equal footing with making progress on Collatz.* The methodology, the contribution split between human and AI, the cross-AI verification strategy, and the now-completed Lean 4 formalization phase are documented in Section 11 and in `METHODOLOGY.md` of the supplementary archive.

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1 Introduction

The Collatz conjecture asserts that the iteration

$$C(n) = \begin{cases} n/2 & \text{if } n \text{ even,} \\ 3n + 1 & \text{if } n \text{ odd,} \end{cases}$$

applied to any positive integer n eventually reaches the trivial cycle $1 \rightarrow 4 \rightarrow 2 \rightarrow 1$. Despite eighty years of attention from combinatorialists, number theorists, and dynamicists, the conjecture remains open. The strongest unconditional progress is due to Tao [1], who established that almost all orbits attain arbitrarily small values relative to the initial point, in a precise density sense.

The bottleneck for a deterministic argument is well known: certain “rebel” starting values produce orbits that climb to many orders of magnitude above their initial value before eventually descending. Empirically these rebels are not isolated curiosities but appear to form structured families which converge in inverse-tree language to common dissipative trunks.

This work proposes that the structural mechanism behind such families is *2-adic shadowing*: rebel positive integers follow, for finite stretches, the trajectory of negative rational fixed points of the Syracuse map under composition. These “phantoms” have no positive integer realization but locally constrain the ν_2 sequence of nearby positive orbits, producing the observed expansive corridors. The same picture has been investigated independently and contemporaneously by Chang [9], who calls the same objects *phantom cycles* and develops a complementary broad-spectrum analysis; we discuss the relationship in detail in Section 1.1 and again in Section 9.1.

We make this framework precise, codify the dynamics as a finite weighted transfer operator on a refined phase quotient, and prove numerically a uniform spectral upper bound which reduces the boundedness of orbits at the critical SCC to an explicit a priori estimate on a family of finite matrices. The fundamental shadowing lemma is mechanically verified in Lean 4 with Mathlib (Section 3.3).

1.1 Related work

The literature on the $3x + 1$ problem is vast; we restrict the discussion to the threads most directly relevant to the framework developed here. Comprehensive bibliographies of earlier work are available in [3, 4, 5].

Parity vectors and congruence theory. The systematic study of parity vectors of the $3x + 1$ map originates with Everett, Terras, and Lagarias in the late 1970s [2]. The classical inverse formula — expressing the residue class of integers realizing a given finite parity sequence — appears in this line and is reformulated explicitly in Rozier [13, Lemma 1] as

$$n \equiv - \sum_{k=0}^{j-1} s_k 2^k 3^{-\sigma_k(S)} \pmod{2^j},$$

for a parity sequence $S = (s_0, \dots, s_{j-1})$ with cumulative 1-counts $\sigma_k(S)$. Lemma 2 below is mathematically equivalent to a particular packaging of this formula in terms of the rational fixed point q_w of the period word; we favor the latter packaging because it is what feeds the episode graph construction in Section 4 and is the form naturally formalized in Lean (Section 3.3).

Non-Archimedean spectral theory. Siegel [10, 11, 12] develops a non-Archimedean spectral theory of the Collatz map via the *Numen* function $\chi_q: \mathbb{Z}_2 \rightarrow \mathbb{Z}_q$ and a Correspondence Principle linking 2-adic values to periodic and divergent orbits of the accelerated Syracuse map. While the underlying 2-adic fixed-point equation $(2^A - 3^L)q_w = C_w$ used here for the phantom representatives coincides with the equation underlying the periodic-point criterion of Bohm–Sontacchi exploited in [10], the present work’s focus on finite transfer operators on a refined phase quotient differs sharply from Siegel’s complex-analytic strategy based on contour integrals and meromorphic continuations of Dirichlet series.

Phantom cycles and transfer operators (concurrent work). The work most closely related to the present one is Chang [9] (*Exploring Collatz Dynamics with Human-LLM Collaboration*), an independently developed Human–LLM collaboration produced through what the author terms a 29-paradigm program comprising several hundred formal results. Three overlapping themes are noteworthy:

1. Chang [9, Def. 7.2] introduces *phantom cycles* of a signature $\sigma = (k_1, \dots, k_\ell)$ as the unique 2-adic fixed point of the corresponding block map F_σ , with explicit equation $(2^K - 3^\ell)\rho = C_\sigma$. This is exactly our q_w construction (Section 3.1) up to notation ($\sigma \leftrightarrow w$, $K \leftrightarrow A$, $\ell \leftrightarrow L$, $\rho \leftrightarrow q_w$).
2. Chang’s 2-adic *repulsion* identity [9, Prop. 7.4], $\nu_2(F_\sigma(x) - \rho) = \nu_2(x - \rho) - K$, is the per-block form of the iterative Lemma 2 below. Chang’s no-prolonged-trapping statement [9, Lemma 9.20] corresponds to Corollary 4.
3. Chang [9, Theorem 7.15] (*Phantom Universality*) establishes that *every* primitive cyclic composition of valuations is a phantom family, a uniform statement strictly stronger than the empirical enumeration we use (Appendix A, $k \leq 24$). The number-theoretic gain rate $R \leq 0.0893$ obtained in [9, Theorem 7.19] is conceptually similar to — but quantitatively distinct from, and computed on a different object than — our spectral bounds; see Section 9.1.

The two works share the core 2-adic phenomenology but differ in strategy: Chang aggregates over all phantom families of growing depth, while the present work concentrates on the single critical SCC with finer phase resolution. Chang’s transfer operator T_K on $(\mathbb{Z}/2^K\mathbb{Z})_{\text{odd}}$ is nilpotent [9, Remark 9.17], whereas our $\text{FULL}_{T,j}$ on the refined phase state $(\nu_2(t), \text{odd}(t) \bmod 4, h \bmod 4)$ has $\rho \approx 0.03$. The fact that the same underlying barrier — a distributional-to-pointwise gap — arises in both formulations independently is, in our view, evidence that this barrier is genuine and not an artifact of either framing.

Operator-theoretic approaches. Berg and Meinardus [15] introduced a linear operator F on the Hardy space $H^2(D)$ and established functional equations for the Collatz iteration. Neklyudov [14] studies the adjoint operator T on the Bergman space, proving hypercyclicity and connecting the index of $\text{Id} - T$ to the number of cycles. Leventides–Poulios [16] employ Koopman operators on ℓ^p spaces and C^* -algebras. Mori [17] formulates the Collatz conjecture in terms of irreducibility of associated C^* -algebras. Each of these works lifts the discrete

Collatz dynamics to an infinite-dimensional linear setting; the present work instead constructs finite-dimensional non-negative matrices from empirical transition data at the critical phantom node.

Spectral methods for transfer operators. The Collatz–Wielandt characterization of the spectral radius of order-preserving homogeneous maps on cones, due to Lemmens, Nussbaum, and collaborators [19, 20], underlies the weighted bound used in Section 6. Keller–Liverani perturbation theory [6] and Baladi’s transfer-operator framework [7] are the natural analytic targets for a possible proof of Conjecture 9 (see Section 10). Recent work on certified spectral approximation of transfer operators [23] and on quasi-compactness of Frobenius–Perron operators [22] is also relevant infrastructure for that direction.

p -adic dynamics and shadowing. Anashin [8] surveys the non-Archimedean theory of discrete systems including Collatz-like maps. Bastos, Caprio, and Messaoudi [21] develop a general theory of shadowing and structural stability for p -adic dynamics, not specialized to Collatz. Laarhoven and de Weger [18] construct modular Collatz graphs on residue classes $\mathbb{Z}/m\mathbb{Z}$ for $m = 2^K$, identifying an isomorphism with binary De Bruijn graphs; our episode graph (Section 4) is a distinct object whose nodes are not residue classes but phantom shadowing episodes.

1.2 Structure of the paper

Section 2 fixes notation for the accelerated Syracuse map and the ν_2 -debt formalism. Section 3 states and proves the exact congruential shadowing lemma; the formalization of this section in Lean 4 is summarized in Section 3.3. Section 4 introduces the episode graph and its critical SCC. Section 5 constructs the finite transfer operator at the critical node. Section 6 develops the weighted Collatz–Wielandt perturbative bound. Section 7 reports numerical evidence in T and j . Section 8 extends the bound to a cross-node operator on the entire SCC. Section 9 formulates the explicit conjectures whose resolution would yield a proof, and includes a quantitative comparison with Chang [9]. Section 10 is an honest accounting of what this work does and does not establish. Section 11 documents the AI-assisted methodology and the contribution split.

2 Setup: accelerated Syracuse map and gravitational debt

For an odd integer n , the *Syracuse step* is

$$S(n) = \frac{3n+1}{2^{\nu_2(3n+1)}}, \quad a(n) := \nu_2(3n+1).$$

We write $a_k = a(n_k)$ for the ν_2 sequence along the orbit $n_0, n_1, n_2, \dots$ with $n_{k+1} = S(n_k)$.

A direct expansion gives, after k Syracuse steps,

$$\log_2 n_k = \log_2 n_0 + k \log_2 3 - \sum_{i=0}^{k-1} a_i + O(1).$$

The dominant term $k \log_2 3 - \sum a_i$ is what we call the *gravitational debt* after k steps:

$$D(k) := k \log_2 3 - \sum_{i=0}^{k-1} a_i.$$

Positive $D(k)$ indicates net expansion to step k ; the orbit height $\log_2 n_k$ tracks $D(k)$ up to bounded fluctuations. The quantity is otherwise classical and is the standard *logarithmic drift* or *total stopping-time deficit* analysis underlying Tao [1] and the historical literature [2, 3, 24]; we use the term “gravitational debt” as informal vocabulary for the program note in Section 11.

The threshold $a_i > \log_2 3 \approx 1.585$ separates dissipative from expansive single steps. The Collatz conjecture is morally equivalent to the assertion that $D(k) \rightarrow -\infty$ along every positive-integer orbit.

3 Phantom rational orbits and the shadowing lemma

3.1 Phantom words and their rational fixed points

Fix a finite word

$$w = (a_0, a_1, \dots, a_{L-1}) \in \mathbb{N}^L, \quad A = A(w) := \sum_{i=0}^{L-1} a_i.$$

The composition $S_w := S_{a_{L-1}} \circ \dots \circ S_{a_0}$ is affine: writing it out one obtains

$$\begin{aligned} S_w(n) &= \frac{3^L n + C_w}{2^A}, & C_0 &= 0, \\ C_{j+1} &= 3C_j + 2^{A_j}, & A_{j+1} &= A_j + a_j. \end{aligned} \tag{1}$$

The fixed point of S_w in $\mathbb{Q}$ is

$$q_w = \frac{C_w}{2^A - 3^L}. \tag{2}$$

The equation (2) is classical: it appears as the “Bohm–Sontacchi” criterion in the Collatz literature [10, Theorem 1] and is a direct consequence of the parity-vector inverse formula of [2, 13]; in the recent literature it is the equation defining the phantom cycles of [9, Def. 7.2], of which the present construction is an instance.

Definition 1 (expansive phantom). The word w is an *expansive phantom* if $3^L > 2^A$. Then $2^A - 3^L < 0$, and assuming $C_w > 0$ (which is automatic from (1)) we have $q_w \in \mathbb{Q}_{<0}$.

The orbit of q_w under S is exactly periodic of length L with ν_2 word w repeated. In the 2-adic completion $\mathbb{Z}_2$, q_w makes sense as a 2-adic integer (since $\gcd(2^A - 3^L, 2) = 1$) and its 2-adic orbit consists of L distinct 2-adic integers cycled by S (lifted to its natural 2-adic extension; see Section 3.3 for the formalized statement).

3.2 Exact congruential shadowing

Let w^∞ denote the right-infinite repetition of w , and let $B_m := \sum_{i=0}^{m-1} a_{i \bmod L}$ for $m \geq 0$ be the partial sum of the periodic word.

Lemma 2 (Exact shadowing). *Let w be an expansive phantom with 2-adic representative q_w , and let n be a positive odd integer. If*

$$n \equiv q_w \pmod{2^{B_m+1}} \quad (\text{equality in } \mathbb{Z}_2 \text{ truncated to } B_m + 1 \text{ bits}),$$

then the first m Syracuse steps of n have ν_2 word exactly $(a_0, a_1, \dots, a_{m-1})$.

In particular, if $n \equiv q_w \pmod{2^{bA+1}}$, the first b periods (i.e. bL Syracuse steps) of n shadow w^∞ .

Remark 3 (relation to existing identities). Lemma 2 is the iterated, step-by-step form of Chang’s per-block 2-adic repulsion identity $\nu_2(F_\sigma(x) - \rho) = \nu_2(x - \rho) - K$ [9, Prop. 7.4], applied L times. Equivalently, it is the inverse parity-vector formula [13, Lemma 1] restated in terms of the rational fixed point q_w as the canonical 2-adic representative of the residue class. The form chosen here is the one that feeds directly into the episode graph construction (Section 4) and into the Lean formalization (Section 3.3).

Proof. We argue by induction on j . Suppose after j Syracuse steps the orbits of n and q_w have followed the same word $(a_0, \dots, a_{j-1})$. Then both have undergone the same affine transformation $S_{(a_0, \dots, a_{j-1})}$, which has multiplier $3^j/2^{B_j}$. Therefore

$$S^j(n) - S^j(q_w) = \frac{3^j}{2^{B_j}}(n - q_w).$$

Taking ν_2 of both sides and using $\nu_2(3^j) = 0$,

$$\nu_2(S^j(n) - S^j(q_w)) = \nu_2(n - q_w) - B_j \geq (B_m + 1) - B_j.$$

For step j we need to extract *exactly* a_j factors of 2 from $3S^j(n) + 1$, which agrees with $3S^j(q_w) + 1$ up to valuation

$$\nu_2(3S^j(n) + 1 - 3S^j(q_w) - 1) = \nu_2(3(S^j(n) - S^j(q_w))) \geq B_m + 1 - B_j.$$

For the next valuation to coincide it suffices that

$$B_m + 1 - B_j \geq a_j + 1, \quad \text{i.e.} \quad B_m \geq B_j + a_j = B_{j+1},$$

which holds for all $j < m$ by definition. Induction completes the proof. The “+1” bit is essential: without it one only ensures agreement modulo 2^{a_j} , not the absence of an additional factor of 2. $\square$

Corollary 4 (No infinite shadowing). *A positive integer cannot shadow w^∞ for all m , since that would force $n = q_w$ in $\mathbb{Z}_2$, hence $n = q_w \in \mathbb{Q}_{<0}$, contradicting $n \in \mathbb{N}$. (Compare [9, Lemma 9.20] for the per-block form.)*

3.3 Formal verification in Lean 4

Lemma 2 and Corollary 4 have been mechanically verified in the Lean 4 theorem prover with Mathlib (Lean v4.29.1, Mathlib v4.29.1). The formalization is `sorry`-free and is included as supplementary material in the project repository under `lean/`. The headline declarations are:

- `CollatzShadowing.exact_shadowing` — Lemma 2, stated paper-faithfully in the 2-adic form: for every positive natural n and every `PhantomWord` w , if $n \equiv q_w \pmod{2^{B_m+1}}$ (encoded as ideal membership in $\mathbb{Z}_2$, see `INVENTORY.md` §1.4), then the first m iterates of n under `Syracuse2adic` produce exactly the periodic word w .
- `CollatzShadowing.exact_shadowing_periods` — the periodic specialization $m = b \cdot L$, with the paper’s congruence bound 2^{bA+1} .
- `CollatzShadowing.no_infinite_period_congruence_expansive` — Corollary 4 for expansive phantoms: no positive natural n can satisfy $n \equiv q_w \pmod{2^{bA+1}}$ for all b .

The supporting infrastructure includes a paper-faithful 2-adic extension of the Syracuse map `Syracuse2adic`: $\mathbb{Z}_2 \rightarrow \mathbb{Z}_2$, the structural `PhantomWord` datatype with positivity constraints, the discharge of the odd-denominator hypothesis on q_w for every phantom, the affine difference identity $S^j(n) - S^j(q_w) = (3^j/2^{B_j})(n - q_w)$ in $\mathbb{Q}_{(2)}$, the strict-ultrametric ν_2 -stability under proximity, and the matching property of q_w 's own S -orbit (so that no auxiliary hypothesis is needed on the user-supplied n beyond the congruence). The total Lean codebase is approximately 2,300 lines across seven files (`Basic`, `Phantom`, `Syracuse2Adic`, `Auxiliary`, `Shadowing`, `NoInfinite`, `INVENTORY`). Build instructions and a complete inventory of the Mathlib API used are documented in `lean/README.md` and `lean/CollatzShadowing/INVENTORY.md` respectively.

The Lean formalization *verifies the shadowing core only*: it does not formalize the episode graph (Section 4), the transfer operator (Section 5), the weighted bound (Section 6), or the conditional reduction (Section 9). Those parts of the paper rest on deterministic numerical computation and on the conjectures stated in Section 9. We discuss this scope explicitly in Section 10 and in Section 11.

4 The episode graph

Lemma 2 shows that an integer n can shadow a phantom w for some maximal number of periods $b = b_w(n)$, after which it must leave the residue class. The natural object for tracking long orbits of integer rebels is therefore not n itself but the sequence of *episodes*

$$n \rightarrow (w_1, b_1) \rightarrow (w_2, b_2) \rightarrow \dots$$

together with terminal events (drops below the starting value).

4.1 The graph

We monitor a finite set of phantoms identified empirically up to $k \leq 24$ (their explicit list is given in Appendix A). For each pair (phantom k , cycle c , depth b) we associate a *node* (k, c, b) . The episode graph G has these nodes as vertices; an edge $(k, c, b) \rightarrow (k', c', b')$ records the empirical observation that some integer orbit shadowed (k, c, b) for exactly b periods, then re-entered shadowing of (k', c', b') .

Remark 5 (scope vs. Phantom Universality). The phantom set used here is *empirical*: it is the list of expansive phantom representatives encountered up to $k \leq 24$ during the dense sweeps of Section 8. By the Phantom Universality theorem of [9, Theorem 7.15], *every* primitive cyclic composition of valuations is a phantom family, so the universe of phantoms is in fact infinite and grows like 1.87^K in $K = \sum k_i$. The empirical sweep finds no SCC outside the listed nodes for the parameter ranges explored, but verifying that no further SCC contributes at higher K is an open question; we discuss this as a missing ingredient of the conditional reduction in Section 10.

4.2 The critical SCC

Empirical exhaustion (script `59_shadow_episode_graph.py`, `60_episode_graph_diagnostics.py`) over the dense sample $b \leq 16$, $t \leq 65535$ identifies a single nontrivial strongly connected component containing

$$(10, 1, b) \text{ for } b \in [1, 15], \quad (11, 1, b) \text{ for } b \in [1, 15], \quad (12, 1, b) \text{ for } b \in [1, 2], \quad (12, 2, b) \text{ for } b \in [1, 2].$$

The *critical node* is $(12, 2, 1)$: it is the unique node of the SCC with significant self-recurrence, and most cross-node transitions terminate at it.

The strategy of the rest of the paper is to prove that this SCC is spectrally subcritical when encoded as a transfer operator with appropriate weights.

5 The critical transfer operator

Fix the critical node $(k, c, b) = (12, 2, 1)$ with phantom word w , period sum A , and 2-adic representative q_w . Every integer n shadowing this node for one period satisfies

$$n \equiv q_w \pmod{2^{A+1}}, \quad \text{i.e.} \quad n = r + t \cdot 2^{A+1}$$

for the unique residue $r = q_w \bmod 2^{A+1}$ and parameter $t \in \mathbb{N}$.

5.1 Refined phase state

For a parameter t , we define the *refined phase state*

$$\sigma(t, h) := (\nu_2(t) \wedge V, \text{odd}(t) \bmod 4, h \bmod 4),$$

where V is a finite valuation cap, $\text{odd}(t) := t/2^{\nu_2(t)}$, and h is the count of monitored hits accumulated so far. The choice of moduli 4 and 4 is the empirical minimum that makes the spectral truncation stable (script `74_phase_transfer_stress.py`).

Remark 6 (difference from Chang’s state space). Chang [9, Sect. 9, Rem. 9.17] works with the transfer operator T_K on the residue space $(\mathbb{Z}/2^K\mathbb{Z})_{\text{odd}}$ and proves it nilpotent: every non-leading eigenvalue vanishes and iterations absorb distributions onto the periodic orbits of the map mod 2^K . The refined phase state above is a *different* quotient: it retains the 2-adic valuation of the lift parameter t (capped at V) and a low-resolution view of the odd part of t together with a hit counter, rather than the full residue mod 2^K . As a consequence $\text{FULL}_{T,j}$ is not nilpotent — it has $\rho(\text{FULL}) \approx 0.03$ at the critical node — and is suitable for a finite spectral bound rather than a finite-step absorption argument.

5.2 Truncated finite operator

For depths $T \in \mathbb{N}$ and $j \in \mathbb{N}$, we define the truncated transfer operator $\text{FULL}_{T,j}$ on the state space

$$\mathcal{S}_{T,j} := \{\sigma(t, h) : t \in [0, 2^T \cdot j), h \in [0, 4)\}$$

as follows. For each source state σ_{src} , enumerate all (t, h) producing it. Trace the Syracuse orbit from $n_0 = r + t \cdot 2^{A+1}$ until either:

1. $n < n_0$ (terminal: contributes no edge);
2. the next monitored hit at the same critical node occurs at $n^* = r + t^* \cdot 2^{A+1}$. This contributes an edge

$$\sigma_{\text{src}} \rightarrow \sigma(t^*, h + 1) \quad \text{with weight} \quad 2^{-\delta}, \quad \delta = \lfloor \log_2 t^* \rfloor - \lfloor \log_2 t \rfloor.$$

We aggregate edges and normalize by source counts:

$$(\text{FULL}_{T,j})_{\sigma, \sigma'} = \frac{1}{|\{(t, h) : \sigma(t, h) = \sigma\}|} \sum_{\substack{(t, h) \\ \sigma(t, h) = \sigma}} \mathbf{1}_{\text{trace lands on } \sigma'} \cdot 2^{-\delta(t)}.$$

By construction $\text{FULL}_{T,j}$ is a substochastic-like nonnegative matrix with row sums ≤ 1 , the deficit corresponding to terminal transitions.

5.3 Empirical signature decomposition

For each pair (r, h) with $r \in [0, 2^T)$, the j lifts

$$t = r + j' \cdot 2^T, \quad j' \in [0, j),$$

produce j traces. We collect the multiset of *transition signatures* (destination state plus δ bit increment) and declare the *core signature* to be the most frequent one. Edges matching this signature are placed in $\text{CORE}_{T,j}$; all other edges are placed in $\text{TAIL}_{T,j}$. By construction

$$\text{FULL}_{T,j} = \text{CORE}_{T,j} + \text{TAIL}_{T,j}.$$

We refer the reader to script `77_high_bit_tail_bound.py` for the implementation details. This decomposition — partitioning by empirical majority of transition signatures — is, to our knowledge, not present in prior work on Collatz transfer operators [9, 14, 16, 17].

6 Weighted perturbative bound

6.1 Collatz–Wielandt with the Perron eigenvector of FULL

The matrices $\text{FULL}_{T,j}$, $\text{CORE}_{T,j}$, $\text{TAIL}_{T,j}$ are nonnegative. Let $v_{T,j}$ denote a right Perron eigenvector of $\text{FULL}_{T,j}$, normalized so $\|v_{T,j}\|_\infty = 1$. By Perron–Frobenius (after a tiny rank-1 perturbation to ensure irreducibility, see remark below), $v_{T,j}$ exists and is strictly positive.

The decomposition $\text{FULL}_{T,j} = \text{CORE}_{T,j} + \text{TAIL}_{T,j}$ acting on $v_{T,j}$ gives, componentwise,

$$(\text{FULL}_{T,j} v_{T,j})_i = (\text{CORE}_{T,j} v_{T,j})_i + (\text{TAIL}_{T,j} v_{T,j})_i.$$

Taking the maximum over i such that $(v_{T,j})_i > 0$ and applying the Collatz–Wielandt characterization of the spectral radius for non-negative matrices [19, 20] gives

$$\boxed{\rho(\text{FULL}_{T,j}) \leq \underbrace{\max_i \frac{(\text{CORE}_{T,j} v_{T,j})_i}{(v_{T,j})_i}}_{\text{core action}} + \underbrace{\max_i \frac{(\text{TAIL}_{T,j} v_{T,j})_i}{(v_{T,j})_i}}_{\text{tail action}}}. \quad (3)$$

We denote the right-hand side $\text{bound}(T, j)$.

Remark 7 (numerical implementation). We compute $v_{T,j}$ by power iteration on $\text{FULL}_{T,j} + \varepsilon J$ with $\varepsilon = 10^{-12}$ and J the all-ones matrix. The rank-1 perturbation removes reducibility numerically without affecting the dominant eigenvalue beyond ε . Convergence is checked in ℓ^∞ norm to tolerance 10^{-12} .

Remark 8 (why CORE’s Perron is wrong). Using the Perron eigenvector of $\text{CORE}_{T,j}$ instead of $\text{FULL}_{T,j}$ yields a divergent “bound”: $\text{CORE}_{T,j}$ is typically reducible, with its Perron eigenvector concentrated on a strict subset of states, and $\text{TAIL}_{T,j}$ routinely sends mass to states outside that subset. The Collatz–Wielandt ratios then blow up. The empirical fix and the theoretical fix both point to using $v_{T,j}$ from $\text{FULL}_{T,j}$.

7 Numerical evidence

7.1 Bound stability in T

We compute $\text{bound}(T, j)$ for $T \in \{8, 10, 12, 14, 16\}$ and $j \in \{16, 32\}$ using script `84_weighted_perturbation_b` (with the eigenvector selection from the previous remark). Results:

T	j	$\rho(\text{FULL})$	core action	tail action	bound
8	16	0.0294	0.0114	0.0294	0.0408
8	32	0.0299	0.0102	0.0299	0.0400
10	16	0.0303	0.0184	0.0303	0.0487
10	32	0.0308	0.0177	0.0308	0.0485
12	16	0.0303	0.0137	0.0303	0.0499
12	32	0.0306	0.0135	0.0306	0.0497
14	16	0.0306	0.0157	0.0306	0.0511
14	32	0.0307	0.0154	0.0307	0.0507
16	16	0.0307	0.0169	0.0307	0.0516

The bound is monotonically increasing in T but the increments decay geometrically:

$$\Delta_8^{10} = +0.0079, \quad \Delta_{10}^{12} = +0.0012, \quad \Delta_{12}^{14} = +0.0012, \quad \Delta_{14}^{16} = +0.0005.$$

A geometric extrapolation yields

$$\lim_{T \rightarrow \infty} \text{bound}(T, 16) \leq 0.054.$$

The bound is essentially independent of j : differences between $j = 16$ and $j = 32$ at fixed T never exceed 0.001.

7.2 Spectral gap $\rho(\text{FULL}) - \rho(\text{CORE})$

Independently, the spectral radii of FULL and CORE are computed (script `83_perturbation_gap_scaling.py`). The gap $\rho(\text{FULL}) - \rho(\text{CORE})$ shrinks with T :

$$T = 6 : 0.025\text{--}0.032, \quad T = 8 : 0.025\text{--}0.028, \quad T = 10 : 0.018\text{--}0.019,$$

$$T = 12 : 0.016\text{--}0.017, \quad T = 14 : 0.015, \quad T = 16 : 0.014.$$

The spectral mass of FULL migrates progressively to CORE as the quotient is refined.

7.3 Truncation stability in j (Cauchy property)

For fixed T , we check whether $\text{FULL}_{T,j}$ admits a Cauchy property in j via two independent metrics (script `85_truncation_stability.py`):

- **signature drift**: the fraction of (r, h) groups whose dominant signature changes between consecutive doublings of j ;
- **bound increment**: $\text{bound}(T, 2j) - \text{bound}(T, j)$.

Results:

T	j	sig_drift_vs_prev	bound	Δ_{prev}
8	16	—	0.0408	—
8	32	0.0078	0.0400	−0.0007
8	64	0.0078	0.0421	+0.0021
8	128	0.0000	0.0425	+0.0004
10	16	—	0.0487	—
10	32	0.0039	0.0485	−0.0003
10	64	0.0088	0.0509	+0.0024
10	128	0.0000	0.0515	+0.0006

Two facts:

1. The signature drift drops to *exactly zero* when j is doubled from 64 to 128, for both $T \in \{8, 10\}$. The empirical transition signatures of all (r, h) groups stabilize.
2. The bound increments decay by a factor ~ 4 per doubling, suggesting a Cauchy bound $\lim_{j \rightarrow \infty} \text{bound}(T = 10, j) \lesssim 0.052$.

Combining the two extrapolations, we conjecture

$$\sup_{T, j} \text{bound}(T, j) < 0.06,$$

a margin of $16\times$ from the criticality threshold 1.

8 Cross-node certificate on the SCC

The single-node bound at the critical node is the worst case: running script `86_scc_node_certificates.py` at $T = 10, j = 32$ across the SCC nodes yields:

node	$\rho(\text{FULL})$	bound	status
(10, 1, 1)	≈ 0	0.0000	OK (return-to-self vacuous)
(10, 1, 2)	≈ 0	0.0000	OK
(11, 1, 1)	≈ 0	0.0000	OK
(11, 1, 2)	≈ 0	0.0000	OK
(12, 1, 1)	0.0035	0.0043	OK
(12, 2, 1)	0.0308	0.0485	OK (worst case)
(12, 2, 2)	0.0007	0.0007	OK

The nodes $(10, 1, *)$, $(11, 1, *)$, $(12, 2, 2)$ have nearly vanishing return-to-self spectrum. They are entry gates to the SCC, not self-recurrent.

8.1 The cross-node operator

We construct the assembled cross-node operator M_{cross} on the union state space

$$\bigsqcup_{(k, c, b) \in \text{SCC}} \mathcal{S}_{T, j}^{(k, c, b)}.$$

Edges are recorded between the source state at one node and the *first hit* state at any node of the SCC, with weight $2^{-\delta_{\text{bit}}}$ where δ_{bit} is the bit-length increment between source n_0 and destination value (a node-invariant quantity). The construction is implemented in script `87_cross_node_transfer.py`.

At $T = 8, j = 8$ on 5 representative SCC nodes:

$$\rho(M_{\text{cross}}) = 0.0366, \quad \text{bound}(M_{\text{cross}}) = 0.0366.$$

The cross-coupling *lowers* the spectrum from the single-node worst case (0.0485). Topologically the SCC is a star centered on $(12, 2, 1)$:

```
(10,1,1) -> (12,2,1)  8128  (gate)
(11,1,1) -> (12,2,1)  8192  (gate)
(12,2,2) -> (12,2,1)  8128  (gate)
```

(12,2,1) -> (12,2,1)	812	(only significant self-loop)
(12,2,1) -> (12,1,1)	308	
(12,1,1) -> (12,1,1)	96	
(12,2,2) -> (12,2,2)	64	
(12,2,1) -> (12,2,2)	8	

The single-node bound was therefore overly conservative; including cross-node transitions strictly improves it.

9 Conjectures and the spectral reduction statement

We isolate the open analytic content as two explicit conjectures.

Conjecture 9 (uniform bound in T). There exists a constant $B_\infty < 1$ such that

$$\sup_{T \in \mathbb{N}, j \in \mathbb{N}} \text{bound}(T, j) \leq B_\infty.$$

Empirically the data support $B_\infty \approx 0.06$.

Conjecture 10 (signature closure in j). For each (T, r, h) there exists $j^*(T, r, h)$ such that the dominant transition signature stabilizes for all $j \geq j^*$, and furthermore

$$\sup_{j \geq j^*} |\text{bound}(T, j) - \text{bound}(T, j^*)| \leq C \cdot 2^{-\beta(j-j^*)}$$

for some $C, \beta > 0$ uniform in (T, r, h) .

Theorem 11 (conditional reduction, sketch). *Assume Conjectures 9 and 10. Then for every positive integer n , the Syracuse orbit $S^k(n)$ satisfies $\liminf_{k \rightarrow \infty} S^k(n) = 1$.*

The argument is the standard one for transfer-operator subcriticality combined with the no-infinite-shadowing Corollary 4: assuming the conjectures, the critical SCC has spectral radius $\leq B_\infty < 1$ on a refined quotient that captures all dynamically relevant phases, hence the return-to-critical mass strictly contracts, hence orbits cannot remain forever near the SCC, hence by exhaustion of the empirical phantom set every orbit eventually escapes to a dissipative regime with average $\nu_2 > \log_2 3$ and descends. The full argument requires an exhaustive enumeration of phantoms below a controlled threshold and a verification that no further SCCs appear; see Section 10 for an honest discussion of the gaps.

What the conditional reduction would buy. The conclusion of Theorem 11, $\liminf_{k \rightarrow \infty} S^k(n) = 1$ for *every* positive integer n , is the full Collatz convergence statement (one direction of the conjecture). This is qualitatively stronger than the unconditional “almost-all” result of Tao [1] — which asserts only logarithmic-density convergence to almost bounded values. The price of the stronger conclusion is precisely the open spectral content of Conjectures 9 and 10: a uniform finite-matrix bound and a Cauchy-type signature closure on the critical SCC. Closing the conjectures unconditionally would therefore push the problem from “almost-all density” to “every n ”, without addressing the trivial-cycle uniqueness question separately treated in the cycle-exclusion literature (e.g. [9], sect. v7 cycle filter).

9.1 Comparison with concurrent work

Chang [9] obtains, by an entirely different route, two quantitative bounds on the per-orbit phantom contribution that are conceptually comparable to the spectral bounds developed here:

- the unconditional spectral bound on the depth-2 extended return kernel [9, Theorem C.3], $\rho(\tilde{B}_2^{\text{ext}}) \leq 5/32 \approx 0.156$, with associated per-return information cost $\geq \log_2(32/5) \approx 2.678$ bits;
- the per-orbit phantom gain rate [9, Theorem 7.19], $R \leq 0.0893 < \varepsilon = 2 - \log_2 3 \approx 0.415$, with safety margin $\varepsilon/R \geq 4.65\times$.

The bounds and ours are computed on different objects, so comparison should be qualitative. Chang’s $\tilde{B}_2^{\text{ext}}$ acts on $(\mathbb{Z}/64\mathbb{Z})_{\text{odd}}$ projected onto the five-element invariant core $I_2 = \{7, 27, 31, 59, 63\} \pmod{64}$, and R aggregates the expanding-drift contribution of every primitive cyclic composition (an exponentially growing phantom universe). Our $\text{FULL}_{T,j}$ acts on a finer phase quotient at a single critical node and yields $\rho(\text{FULL}) \approx 0.0308$ at $T = 16, j = 32$ with weighted upper bound $\text{bound}(T, j) \leq 0.0516$. The cross-node operator M_{cross} further sharpens the SCC-level estimate to 0.0366.

quantity	value	object
$\rho(\text{FULL})$, this work, $T = 16, j = 32$	0.0308	critical-node single-node
$\text{bound}(T, j)$, this work	≤ 0.0516	weighted Collatz–Wielandt
$\rho(M_{\text{cross}})$, this work	0.0366	cross-node on SCC
R , Chang [9, Thm. 7.19]	≤ 0.0893	per-orbit phantom gain
$\rho(\tilde{B}_2^{\text{ext}})$, Chang [9, Thm. C.3]	$\leq 5/32 \approx 0.156$	depth-2 I_2 kernel

The two programs differ in coverage and granularity. Chang’s bounds are unconditional and aggregate but coarser; ours are finer and local but conditional on the conjectures of this section. The fact that two independently developed AI-assisted programs arrive, by very different routes, at quantitatively similar bounds is, in our view, evidence that the underlying phenomenology is robust.

Scale of parameters explored. The two analyses operate at very different parameter scales, and the comparison should be read with this in mind. Chang’s exact finite computation of the per-orbit gain rate [9, Theorem 7.19, Step 2] sums $R(K)$ for $K = 3, \dots, 500$ via Möbius inversion of the primitive cyclic-composition counts $M(K, \ell)$, with an analytic Chernoff–Cramér tail beyond $K_0 = 500$, aggregating over a phantom universe whose size grows like 1.87^K . The numerical evidence developed here covers $T \leq 16$ and $j \leq 128$ at the single critical node $(12, 2, 1)$, with full phase-state resolution $(\nu_2(t), \text{odd}(t) \bmod 4, h \bmod 4)$. The two regimes are complementary — broad-shallow aggregation over an exponentially growing family count versus narrow-deep resolution at a single node — and not directly substitutable. Extending the present sweep to higher T is computationally feasible (the matrix sizes scale polynomially in T and linearly in j) and is part of the planned program for a future revision; we note, however, that the geometric decay of bound increments observed through $T = 16$ already suggests a finite limit, so the principal analytic content lies in proving that limit, not in extending the sweep numerically.

The same barrier in two formulations. Conjectures 9 and 10 are a finite-matrix re-expression of the same distributional-to-pointwise gap that [9, Sect. 9] identifies as the irreducible obstruction to closing every standard route. Concretely, Chang’s gain-observable recurrence [9, Conj. 9.18],

$$\eta_{K+3}(n_0) \leq \lambda \eta_K(n_0) + CK^{-\beta},$$

is a depthwise sharpening of the same kind of bound that Conjecture 9 requires uniformly in T . We do not claim that closing one closes the other — the operators and observables are

different — but the structural barrier is the same. We therefore adopt the position of [9]: *the present work is not a proof of the Collatz conjecture; it characterizes why a particular structural reduction does not close unconditionally.*

10 What this is and what this is not

This work is a *program*, not a proof. We summarize what is established, what is empirical, and what is missing.

Established (rigorously, with mechanical verification).

- Lemma 2 (exact congruential shadowing) is a theorem with elementary proof. It identifies the precise residue condition for b -fold phantom shadowing. *It has been mechanically verified in Lean 4 with Mathlib (Section 3.3).*
- Corollary 4 (no infinite shadowing) follows formally and has also been verified in Lean 4 in the form `no_infinite_period_congruence_expansive`.
- The construction of the finite transfer operators $\text{FULL}_{T,j}$, $\text{CORE}_{T,j}$, $\text{TAIL}_{T,j}$ at the critical node is a deterministic algorithm; the matrices are exactly computable for all finite T, j .
- The bound (3) is a Collatz–Wielandt consequence of nonnegativity [19]; it gives a true upper bound on $\rho(\text{FULL}_{T,j})$ for each finite (T, j) .

Empirical (numerically supported, not proved).

- All numerical bounds reported in Sections 7 and 8.
- The geometric decay of bound increments in T and j .
- The closure of dominant signatures at $j = 128$ for $T \in \{8, 10\}$.
- The identification of $(12, 2, 1)$ as the worst-case node of the SCC.
- The structural “star” topology of the SCC around $(12, 2, 1)$.

Missing.

- Proofs of Conjectures 9 and 10. We believe the natural framework is transfer-operator analysis on a suitable Banach space of 2-adic Lipschitz functions; a Lasota–Yorke inequality for the family $\{\text{FULL}_T\}$, combined with Hennion’s theorem and Keller–Liverani perturbation theory [6], would yield Conjecture 9. We have not carried this out, and recent literature on quasi-compactness [22] and certified spectral approximation [23] suggests this is a non-trivial analytic project. The shared barrier with [9, Sect. 9] (the distributional-to-pointwise gap) suggests that closing it conditionally on a single operator may itself be hard.
- Verification that the empirical phantom set ($k \leq 24$) is complete in the sense that no SCC outside the listed nodes contributes. By [9, Theorem 7.15], the underlying phantom universe is infinite; verifying that no further SCC contributes *at the dynamical level* for orbits below an explicit threshold is open. Empirical sweeps to $b \leq 16$, $t \leq 65535$ find none, but this is not a proof.

- An explicit taxonomic mapping of the empirical phantom set used here ($k = 10, 11, 12$ cycle 1, 12 cycle 2, 20, see Appendix A) onto the universe of primitive cyclic compositions enumerated in [9, Theorem 7.15]. This is a finite enumeration — one would tabulate, for each depth K up to a chosen cutoff (say $K \leq 12$ or $K \leq 16$), all primitive compositions $(k_1, \dots, k_\ell)$ with expansive drift $\Delta = \ell \log_2 3 - K > 0$, identify the ones realised by our four monitored phantoms, and verify whether any further primitive composition at the same depths produces an SCC of the episode graph not already covered. We have not carried out this enumeration in the present revision and defer it to subsequent work; until it is done, the empirical scope of the SCC analysis in Section 8 is, strictly, a working hypothesis.
- A formal version of Theorem 11. Lemma 2 and Corollary 4 are formalized in Lean (Section 3.3); a natural next phase of the program is to formalize the episode graph and the transfer operator construction. We note explicitly that the *conjectures* themselves (Conjectures 9 and 10) involve infinite suprema and asymptotic decay, both of which are statable but presently unprovable in current Mathlib.

Honest claim. We have reduced the open structural content of the Collatz conjecture, at the level of the empirical critical SCC, to a problem of *spectral perturbation theory on a finite explicit family of nonnegative matrices*, with the foundational shadowing lemma mechanically verified, supported by numerical evidence with a $16\times$ safety margin. We do not claim the conjecture itself.

10.1 Planned next steps

The work above isolates several finite, well-defined targets whose resolution would either tighten the present reduction or formalize parts of it that currently rest on numerical evidence. We list them in increasing order of analytic difficulty.

Priority A — Phantom-set taxonomy with respect to [9, Theorem 7.15]. The present empirical phantom set ($k \leq 24$, Appendix A) is small relative to the universe of primitive cyclic compositions enumerated by Chang. A finite but non-trivial enumeration is in scope: for a chosen depth cutoff K_0 (say $K_0 = 16$ or 20), enumerate all primitive cyclic compositions $(k_1, \dots, k_\ell)$ with expansive drift $\Delta = \ell \log_2 3 - K > 0$, compute their 2-adic representatives q_w , simulate sufficiently many integer orbits shadowing them, and verify either that every resulting strongly connected component is already covered by the SCC of Section 8, or otherwise produce explicit new SCCs to be added. The exercise is bounded and scriptable, but the Möbius enumeration of primitive necklaces grows rapidly with K_0 and the per-composition orbit simulation is non-trivial; we estimate one to two weeks of focused scripting effort before any honest claim of completeness can be made. This is the first target we intend to address.

Priority B — Lean formalization of the episode graph and the transfer operator. The Lean 4 + Mathlib formalization of Section 3.3 covers the shadowing core (Lemma 2 and Corollary 4) only. A natural extension — conditional on Priority A having fixed the working phantom set — is to formalize the episode graph itself: introduce a `SimpleDigraph` on the nodes (k, c, b) , formalize the SCC computation (e.g. via Tarjan’s algorithm or an existing Mathlib graph-theoretic construction), define the truncated transfer operator $\text{FULL}_{T,j}$ and the decomposition $\text{FULL}_{T,j} = \text{CORE}_{T,j} + \text{TAIL}_{T,j}$, and prove that the weighted Collatz–Wielandt bound (3) is a true upper bound on $\rho(\text{FULL}_{T,j})$ for each finite (T, j) . The work is purely combinatorial and does not require new mathematics, but the Mathlib graph and

matrix infrastructure required is non-trivial; a realistic estimate is two to four weeks of LLM-assisted Lean sessions, comparable to the scope of the current Lean phase.

Priority C — Spectral-gap analysis toward Conjecture 9. The strongest analytic target, and the most uncertain, is to prove a *partial* bound on the family $\{\text{FULL}_T\}_{T \geq T_0}$ that does not require full closure of Conjecture 9. The natural tools are a Lasota–Yorke inequality on a Banach space of 2-adic Lipschitz functions, Hennion’s theorem to extract a spectral gap, and Keller–Liverani perturbation theory [6] to control the dependence on T . Recent infrastructure on quasi-compactness of transfer operators with countable branches [22] and certified spectral approximation [23] provides a starting point. We do not have the analytic expertise to carry out this program in isolation and consider it the most natural target for collaboration with a researcher working in the area; a single concrete intermediate result, short of closing Conjecture 9, would already be a meaningful step forward.

We intend to address these in the listed order: A first (scriptable, contained), then B (Lean infrastructure extension, contained), then C (open-ended, collaboration-dependent).

11 Methodology and AI disclosure

This paper is the first artifact of a personal research program by the author, an independent researcher, exploring a question of methodology as much as one of mathematics:

To what extent can sustained collaboration with general-purpose large language model assistants enable an independent researcher to make non-standard, well-founded, externally reviewable attempts at open mathematical problems?

The question is not a rhetorical one. It is the experiment of which the present manuscript — and the parallel Lean formalization described in Section 3.3 — are concrete artifacts. **Verifying what such collaboration can and cannot deliver, in a setting where each step is reproducible and openly reviewable, is an explicit secondary goal of this work**, on equal footing with the substantive Collatz progress.

11.1 Why this disclosure matters

A clear separation of contributions is essential for the reader’s correct evaluation of the material. Errors that would normally be caught by departmental colleagues, weekly seminars, or referee comments are not part of the development cycle here; they have to be substituted by other safeguards (numerical reproducibility, mechanical verification, multi-AI cross-checking, public release for external scrutiny).

11.2 Author’s contribution

The conceptual framing is due to the author: the gravitational debt metaphor for orbit expansion, the empirical observation that $\nu_2 = 1$ corridors are the locus of explosive growth, the structural intuition that integer rebels shadow 2-adic phantom orbits associated with negative rational fixed points of the Syracuse map, the strategic pivot from the empirical quasi-Lyapunov candidates of the early phase to the present spectral program (after their false-alarm pattern proved structurally inseparable from genuine rebels), and the decision to publish at this stage. The author led the research direction throughout: the choices of what to investigate, what to refine, when to stop scaling, and what to publish are the author’s. The author is also responsible for the literature reconnaissance phase that produced the related-work analysis of Section 1.1.

11.3 AI assistance, by phase

Early empirical phase (`scripts/early_empirical/`). Scripts numbered roughly 01 through 48, plus `validate_*.py`. Topics: orbit profiling, debt accumulation metrics, family identification in inverse trees, false-alarm classification, empirical search for the candidate quasi-Lyapunov H function. Developed with assistance from **OpenAI Codex** and **Google Gemini**. The author guided the empirical questions and validated the results manually against worked examples.

Spectral program phase (`scripts/spectral_program/`). Scripts numbered 49 through 87. Topics: phantom rational representatives, shadowing congruence verification, episode graph construction, strongly connected component identification, the critical symbolic operator, the CORE/TAIL decomposition, the weighted Collatz–Wielandt bound, truncation stability, per-node and cross-node SCC certificates. These scripts, the mathematical formalization of Lemma 2 and the weighted Collatz–Wielandt bound, the construction of the transfer operator, the conditional reduction theorem, and the entire $\text{\LaTeX}$ manuscript were developed with substantial assistance from **Anthropic’s Claude**, accessed via the Claude Code interface in extended interactive sessions.

In particular, the following content originated in conversation with Claude:

- the proof of Lemma 2 (induction on the bit valuation of the difference $S^j(n) - S^j(q_w)$);
- the construction of the refined phase state $(\nu_2(t), \text{odd}(t) \bmod 4, h \bmod 4)$;
- the empirical core/tail signature decomposition;
- the derivation of the weighted Collatz–Wielandt bound and the observation that the right Perron eigenvector of FULL (rather than CORE) is the correct test vector;
- the cross-node operator construction and the topological interpretation of the SCC as a star centered on $(12, 2, 1)$;
- the formulation of the open conjectures and the conditional reduction theorem.

The author’s role in this phase was to ask follow-up questions, to suggest experiments, to validate against numerical sanity checks, and to make strategic decisions about scope and publication timing.

Lean formalization phase (`lean/`). The Lean 4 + Mathlib formalization of Lemma 2 and Corollary 4 (Section 3.3) was carried out in a sequence of focused sessions across approximately three calendar days, alternating **Claude** (Claude Code interface) and **OpenAI Codex** as primary technical assistants. Cross-AI checking was used to identify mistakes that one assistant did not catch: in particular, several Mathlib API mismatches arising from the rapid evolution of Mathlib lemma names (e.g. deprecated `push_neg`, the maximal-ideal API renaming, `List.get?` removal in v4.29.1) were identified by one assistant and fixed by the other. The author’s role was to set session-level goals, validate that each session’s claimed artifact was honest (zero `sorry`, `lake build clean`, statements faithful to the paper’s language), and to maintain the operational `TODO.md` across sessions. The session log at the bottom of `lean/TODO.md` records the exact sequence of steps and which assistant produced which artifact.

Literature reconnaissance phase (arxiv-affini/, separate from the project repository). The related-work content of Section 1.1 is the output of a separate AI-assisted reconnaissance: a corpus of ~ 100 arXiv papers in math.NT, math.DS, math.SP, math.FA, and related categories was assembled, ranked by topical proximity to the present work, and reviewed for explicit overlaps and differences. The author validated the most consequential candidates (Chang [9], Siegel [10], Rozier [13]) by direct inspection of the relevant sections, not only via abstracts. The reconnaissance corpus and its automated ranking scripts are not committed to the public project repository: they consist largely of third-party PDFs and of intermediate scratch documents whose contents are subsumed by Section 1.1 of this paper. The methodology of the reconnaissance is, however, openly documented; see the “Literature reconnaissance method” note appended to `METHODOLOGY.md`.

11.4 Cross-AI verification protocol

The methodology that emerged in practice, and which we recommend for analogous attempts, is a four-stage verification cycle for every non-trivial mathematical claim:

1. the claim is first developed in dialogue with a primary AI assistant;
2. it is then re-examined, in cold-start form, by a second AI assistant whose role is adversarial review;
3. where applicable, it is reduced to a deterministic numerical experiment whose output can be inspected against worked examples by the author;
4. where applicable, it is mechanically verified in Lean.

Stages 3 and 4 are the strongest safeguards: a numerical experiment that fails to reproduce a claimed identity exposes the error immediately, and a Lean proof that fails to type-check exposes any logical gap that the dialogue rounded over.

11.5 Verification status, current

- All numerical results in Sections 7 and 8 are deterministic and reproducible from the supplementary scripts; anyone with Python 3.10+, NumPy, and SciPy can re-run them.
- Lemma 2 and Corollary 4 are mechanically verified in Lean 4 with Mathlib; the project is sorry-free. See Section 3.3 and `lean/`.
- As of this version, the work has not been reviewed by a human mathematical expert. The construction of the transfer operator, the weighted bound, and the conditional reduction theorem rest on AI-assisted formalization that has not yet been externally validated. External expert review is invited.
- The literature reconnaissance is necessarily limited by the corpus consulted (~ 100 papers up to and including early 2026). Genuinely related work outside this corpus is possible; we discuss this risk and a mitigation plan in Section 11.6 below.

11.6 Future search for analogous studies

The risk of inadvertent duplication is highest in the intersection of (i) symbolic / p -adic dynamics, (ii) finite transfer operators, (iii) Collatz-like iterations. The discovery of [9] two months after our v1 was deposited is a concrete instance of this risk. To reduce it for future iterations of this work and for adjacent problems, the author commits to the following recurring search procedure (also documented in `METHODOLOGY.md`):

1. monthly arXiv listing scans in `math.NT`, `math.DS`, `math.SP`, `math.FA`, `math.CO`, with manual screening of titles and abstracts for the keywords *phantom*, *shadow*, *transfer operator*, *Collatz*, *Syracuse*, $3x + 1$, *2-adic*, *spectral gap*, *Collatz–Wielandt*, *Perron*;
2. targeted Google Scholar and Semantic Scholar queries with the same keyword combinations, plus citation-graph traversal of the seed papers [9, 10, 13, 14, 16, 19];
3. AMS MathSciNet and zbMATH searches under MSC primary 11B83 and 37N99, with secondary 47A10 and 11S99;
4. monitoring of the arXiv 2603.* and 2604.* releases that cite the seed papers above;
5. should any future related work surface, an explicit addendum or revised version note will be prepared and added to the public repository within a reasonable time-frame.

The author considers this duty of continuous reconnaissance an intrinsic part of an AI-assisted research program: standard in-person community signals (department seminars, ad-hoc discussions, workshop announcements) that ordinarily flag related work for institutionally embedded researchers are absent here, so the surveillance has to be made explicit.

11.7 How to engage

- **For the math.** Read Sections 3, 5, and 6 of this paper. If you find an error, an unjustified step, or an implicit assumption worth flagging, please contact the author at the address on the title page.
- **For the Lean formalization.** Clone the repository and run `lake build` from the `lean/` directory. The headline declarations (`exact_shadowing`, `exact_shadowing_periods`, `no_infinite_period_congruence_expansive`) compile without `sorry` on Lean v4.29.1 and Mathlib v4.29.1.
- **For the numerics.** Clone the repository, install NumPy and SciPy, and re-run any of the scripts in `scripts/spectral_program/`.
- **For collaboration.** The author is open to collaboration with anyone willing to take the proofs seriously, and especially welcomes (a) attempts to close Conjecture 9 via a Lasota–Yorke / Keller–Liverani argument, and (b) independent verification of the SCC enumeration of Section 8.

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A Phantom records used in this work

The expansive phantom representatives identified empirically up to $k = 24$ and used as monitored nodes in Section 4:

k	$q_w \approx$
10	$-5\,739\,192\,681\,529/2\,404\,426\,874\,857 \approx -2.387$
11	$-2\,199\,012\,694\,031/709\,849\,655\,971 \approx -3.098$
12 (cycle 1)	$-2123/1675 \approx -1.267$
12 (cycle 2)	$-817/601 \approx -1.359$
20	$-104\,312\,644\,103/30\,307\,317\,785 \approx -3.442$

The full list and the construction of these representatives from the rational fixed point formula (2) is implemented in `54_phantom_rational_shadows.py`. By [9, Theorem 7.15] this list is a small subset of the infinite phantom universe; we use it as a finite working set for the empirical SCC analysis of Section 8.

B Supplementary scripts and Lean code

All supporting material is available in the GitHub repository

<https://github.com/PieroBorgatta/Collatz>

The principal Python scripts and what each computes:

54_phantom_rational_shadows.py

Computes the 2-adic representatives of expansive phantom cycles; underpins all subsequent constructions.

55_shadowing_congruence.py

Verifies Lemma 2 on the empirical phantom set for $b \leq 10$.

59_shadow_episode_graph.py, 60_episode_graph_diagnostics.py

Constructs the episode graph and identifies its SCCs.

75_critical_symbolic_operator.py

Builds the truncated finite operator at the critical node.

77_high_bit_tail_bound.py

Decomposes $\text{FULL} = \text{CORE} + \text{TAIL}$ from the empirical signature majority.

82_boundary_scc_spectral_scaling.py

Spectral scaling on SCCs touching the boundary layer $\nu_2(t) \geq T$.

83_perturbation_gap_scaling.py

Measures the gap $\rho(\text{FULL}) - \rho(\text{CORE})$.

84_weighted_perturbation_bound.py

Computes the weighted bound (3).

85_truncation_stability.py

Cauchy/signature stability of the truncation in j .

86_scc_node_certificates.py

Per-node bounds on the SCC.

87_cross_node_transfer.py

Assembled cross-node transfer operator on the SCC.

The Lean 4 + Mathlib formalization is in the subdirectory **lean/** of the repository:

lean/CollatzShadowing/Basic.lean

The accelerated Syracuse map $S: \mathbb{N} \rightarrow \mathbb{N}$ and the valuations ν_2 on $\mathbb{N}, \mathbb{Z}, \mathbb{Q}, \mathbb{Z}_2$.

lean/CollatzShadowing/Phantom.lean

The PhantomWord datatype, affine fold of the Syracuse step, the named coefficients C_w, A_w , the rational fixed point q_w , and its $\mathbb{Z}_2$ representative.

lean/CollatzShadowing/Syracuse2Adic.lean

The total 2-adic extension $\text{Syracuse2adic}: \mathbb{Z}_2 \rightarrow \mathbb{Z}_2$ and the bridge to S on $\mathbb{N}$.

lean/CollatzShadowing/Auxiliary.lean

Supporting lemmas: $\nu_2(3n) = \nu_2(n)$ on $\mathbb{Z}_2$, odd-denominator of q_w , the affine difference identity $S^j(n) - S^j(q_w) = (3^j/2^{B_j})(n - q_w)$, the strict ultrametric stability of ν_2 under proximity, the matching property of q_w 's own orbit, and periodicity identity $B(bL) = bA$.

lean/CollatzShadowing/Shadowing.lean

Lemma 2 (`exact_shadowing`) and its periodic specialization (`exact_shadowing_periods`); both sorry-free.

lean/CollatzShadowing/NoInfinite.lean

Corollary 4 (`no_infinite_period_congruence_expansive`); sorry-free.

lean/CollatzShadowing/INVENTORY.md

A complete inventory of the Mathlib API used and of the chosen modeling decisions (in particular, congruence as ideal membership in $\mathbb{Z}_2$).

Building requires Lean v4.29.1 and Mathlib v4.29.1, both pinned in the `lean-toolchain` and `lakefile.toml` files.